

DSA0405 - FUNDAMENTALS OF DATA SCIENCE

CO3AT1 – CASE STUDY

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Case Study 6 – Detecting an Unusual Customer

Question:

A shopping website finds that the average amount spent by customers is **₹2,000**, with a **standard deviation of ₹400**. A particular customer spends **₹3,200**.

1. Calculate the customer's z-score.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of **-1.5**, what does it tell you?

Answer

Introduction

A **z-score** measures how far a value is from the mean in terms of standard deviations. It helps identify whether a value is typical or unusual. Businesses use z-scores to analyze customer spending patterns and detect outliers.

Given Data

- Mean (μ) = ₹2,000
- Standard Deviation (σ) = ₹400
- Customer Spending (X) = ₹3,200

Formula

$$Z = \frac{X - \mu}{\sigma}$$

1. Calculate the Customer's Z-score

$$Z = \frac{3200 - 2000}{400} = \frac{1200}{400} = 3$$

Answer: Z-score = 3

The customer's spending is **3 standard deviations above the average**.

2. Is the Customer's Spending Unusually High?

Yes.

A z-score greater than **+2** usually indicates an unusually high value. Since the customer's z-score is **3**, the spending is much higher than that of an average customer.

The company may consider this customer a **high-value customer** and offer special rewards or personalized promotions.

3. What Does a Negative Z-score Mean?

A negative z-score means the customer's spending is **below the average**.

For example, if a customer has a z-score of **-1**, they spend one standard deviation less than the average customer. It does not indicate an error; it simply shows lower spending.

4. Customer with Z-score = -1.5

Using the formula:

$$\begin{aligned} X &= \mu + Z\sigma \\ X &= 2000 + (-1.5 \times 400) = 2000 - 600 = ₹1400 \end{aligned}$$

Answer: The customer spends **₹1,400**, which is **below the average** but not extremely low.

Applications of Z-score

- Detects unusual customer spending.
- Identifies high-value customers.
- Helps in fraud detection.
- Supports customer segmentation.

- Used in machine learning for data normalization.

Conclusion

The customer who spends **₹3,200** has a **z-score of 3**, indicating unusually high spending compared to the average of **₹2,000**. A **negative z-score** represents spending below the average. Z-scores help businesses understand customer behavior, identify outliers, and make better marketing and business decisions.

Case Study 7 – Delivery Time Analysis

Question:

A delivery company records delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the mean and median.
 2. Find Q1 and Q3.
 3. Calculate the IQR.
 4. Check for outliers using the $1.5 \times \text{IQR}$ rule.
 5. Explain whether the mean or median gives a better description of the typical delivery time.
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Answer

Introduction

Measures such as **mean, median, quartiles, and interquartile range (IQR)** help analyze data and identify unusual values. In this case, they are used to study delivery times and determine whether any delivery is significantly different from the rest.

Given Data

Delivery times (minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Number of observations (**n**) = **10**

1. Calculate the Mean and Median

Mean

$$\begin{aligned}\text{Mean} &= \frac{25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60}{10} \\ &= \frac{326}{10} = 32.6 \text{ minutes}\end{aligned}$$

Mean = 32.6 minutes

Median

Since there are 10 values (even number), the median is the average of the 5th and 6th values.

5th value = **30**

6th value = **30**

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Median = 30 minutes

2. Find Q1 and Q3

Lower half:

25, 27, 28, 29, 30

Middle value = **28**

Q1 = 28

Upper half:

30, 31, 32, 34, 60

Middle value = **32**

Q3 = 32

3. Calculate the IQR

$$\begin{aligned} IQR &= Q3 - Q1 \\ &= 32 - 28 = 4 \end{aligned}$$

IQR = 4

4. Check for Outliers (1.5 × IQR Rule)

Lower Limit

$$\begin{aligned} Q1 - 1.5(IQR) \\ = 28 - 1.5(4) = 28 - 6 = 22 \end{aligned}$$

Upper Limit

$$\begin{aligned} Q3 + 1.5(IQR) \\ = 32 + 6 = 38 \end{aligned}$$

Any value **below 22** or **above 38** is an outlier.

The dataset contains **60**, which is greater than **38**.

Therefore, 60 minutes is an outlier.

5. Mean or Median – Which is Better?

The **median** is a better measure because the value **60** is an outlier that increases the mean.

- Mean = **32.6 minutes**
- Median = **30 minutes**

Most deliveries are completed in **25–34 minutes**, so the median better represents the typical delivery time.

Applications

- Measures delivery performance.
 - Identifies unusually delayed deliveries.
 - Helps improve logistics and route planning.
 - Assists managers in monitoring service quality.
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Conclusion

The average delivery time is **32.6 minutes**, while the median is **30 minutes**. The quartiles are **Q1 = 28** and **Q3 = 32**, giving an **IQR of 4**. Using the **$1.5 \times \text{IQR}$ rule**, **60 minutes** is identified as an outlier. Therefore, the **median** is a better measure of the typical delivery time because it is not affected by the unusually long delivery.

Case Study 8 – Two Cities' Temperatures

Question:

The average daily temperatures (°C) for 7 days are:

- **City A:** 28, 29, 30, 30, 31, 30, 32
 - **City B:** 20, 25, 30, 35, 30, 25, 45
1. Calculate the mean temperature for each city.
 2. Calculate the variance for each city.
 3. Which city has greater temperature variability?
 4. Why can two datasets have the same mean but different standard deviations?
 5. Which city has more predictable temperatures?
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Answer

Introduction

The **mean** shows the average value of a dataset, while the **variance** measures how much the values differ from the mean. A lower variance indicates more consistent data, whereas a higher variance shows greater variation. In this case, we compare the temperature patterns of two cities.

Given Data

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

Number of observations (**n**) = 7

1. Calculate the Mean Temperature

City A

$$\text{Mean} = \frac{28 + 29 + 30 + 30 + 31 + 30 + 32}{7} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City A = 30°C

City B

$$\text{Mean} = \frac{20 + 25 + 30 + 35 + 30 + 25 + 45}{7} = \frac{210}{7} = 30^{\circ}\text{C}$$

Mean of City B = 30°C

2. Calculate the Variance

City A

Temperature (x - 30)²

28 4

29 1

30 0

30 0

31 1

30 0

32 4

Sum = **10**

$$\text{Variance} = \frac{10}{7} = 1.43$$

Variance of City A = 1.43

City B

Temperature (x - 30)²

20 100

25 25

30 0

35 25

30 0

25 25

45 225

Sum = **400**

$$\text{Variance} = \frac{400}{7} = 57.14$$

Variance of City B = 57.14

3. Which City Has Greater Temperature Variability?

City **B** has a much higher variance (**57.14**) than City **A** (**1.43**).

Therefore, **City B has greater temperature variability**, as its temperatures change more from day to day.

4. Why Can Two Datasets Have the Same Mean but Different Standard Deviations?

The **mean** only represents the average value, while the **standard deviation** measures how spread out the data is.

In this example, both cities have the same average temperature of **30°C**, but City A's temperatures stay close to 30°C, whereas City B's temperatures vary widely from **20°C to 45°C**. Hence, both datasets have the same mean but different standard deviations.

5. Which City Has More Predictable Temperatures?

City A has more predictable temperatures because its variance is much lower. The temperatures remain close to the average throughout the week, making the weather more stable and consistent.

Conclusion

Both cities have the **same mean temperature of 30°C**, but their temperature patterns are different. **City A** has a **variance of 1.43**, indicating stable temperatures, while **City B** has a **variance of 57.14**, showing large fluctuations. Therefore, **City B has greater variability**, and **City A has more predictable temperatures**.

Case Study 9 – Data Center CPU Usage

Question:

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate the mean, median, and mode.
 2. Calculate the range.
 3. Find Q1, Q3, and IQR.
 4. Identify possible outliers.
 5. Determine the direction of skewness.
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Answer

Introduction

Measures such as **mean, median, mode, range, quartiles, and interquartile range (IQR)** help summarize a dataset and identify unusual values. In this case, they are used to analyze CPU usage and determine whether the data contains any outliers or skewness.

Given Data

CPU Usage (%):

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Number of observations (**n**) = **10**

1. Calculate the Mean, Median, and Mode

Mean

$$\text{Mean} = \frac{40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90}{10} = \frac{513}{10} = 51.3\%$$

Mean = 51.3%

Median

Since there are 10 observations, the median is the average of the 5th and 6th values.

$$\text{Median} = \frac{46 + 48}{2} = 47\%$$

Median = 47%

Mode

The value **45** occurs twice, while all other values occur only once.

Mode = 45%

2. Calculate the Range

$$\begin{aligned}\text{Range} &= \text{Maximum Value} - \text{Minimum Value} \\ &= 90 - 40 = 50\end{aligned}$$

Range = 50%

3. Find Q1, Q3, and IQR

Lower half:

40, 42, 45, 45, 46

Middle value = **45**

Q1 = 45

Upper half:

48, 50, 52, 55, 90

Middle value = **52**

Q3 = 52

IQR

$$IQR = Q3 - Q1 = 52 - 45 = 7$$

IQR = 7

4. Identify Possible Outliers

Lower Limit

$$Q1 - 1.5(IQR) = 45 - 1.5(7) = 45 - 10.5 = 34.5$$

Upper Limit

$$Q3 + 1.5(IQR) = 52 + 10.5 = 62.5$$

Any value **below 34.5** or **above 62.5** is considered an outlier.

The value **90** is greater than **62.5**, so it is an **outlier**.

5. Determine the Direction of Skewness

The dataset contains a very high value (**90**), which increases the mean.

- **Mean = 51.3%**
- **Median = 47%**

Since the **mean is greater than the median**, the distribution is **positively skewed (right-skewed)**.

This means that most CPU usage values are clustered between **40% and 55%**, while one unusually high value stretches the distribution to the right.

Conclusion

The CPU usage data has a **mean of 51.3%**, **median of 47%**, and **mode of 45%**. The **range is 50%**, and the quartiles are **Q1 = 45** and **Q3 = 52**, giving an **IQR of 7**. Using the **1.5 × IQR rule**, **90%** is identified as an **outlier**. The data is **positively (right) skewed** because the high CPU usage value increases the mean above the median.

Case Study 10 – Comparing Two Datasets

Question:

Two data analysts receive the following datasets:

- **Dataset A:** 10, 20, 30, 40, 50
 - **Dataset B:** 28, 29, 30, 31, 32
1. Calculate the mean of both datasets.
 2. Calculate the variance of both.
 3. Calculate the standard deviation of both.
 4. Which dataset has greater spread?
 5. If both datasets represent model errors, which model appears more consistent?
 6. Explain why mean alone is not enough to compare datasets.
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Answer

Introduction

The **mean** gives the average value of a dataset, while the **variance** and **standard deviation** measure how much the data values differ from the mean. Two datasets can have the same mean but very different levels of variation. Therefore, measures of spread are important when comparing datasets.

Given Data

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

Number of observations (**n**) = 5

1. Calculate the Mean

Dataset A

$$\text{Mean} = \frac{10 + 20 + 30 + 40 + 50}{5} = \frac{150}{5} = 30$$

Mean of Dataset A = 30

Dataset B

$$\text{Mean} = \frac{28 + 29 + 30 + 31 + 32}{5} = \frac{150}{5} = 30$$

Mean of Dataset B = 30

2. Calculate the Variance

Dataset A

Value (x - 30)²

10 400

20 100

30 0

40 100

50 400

Sum = **1000**

$$\text{Variance} = \frac{1000}{5} = 200$$

Variance of Dataset A = 200

Dataset B

Value (x - 30)²

28 4

29 1

30 0

31 1

32 4

Sum = **10**

$$\text{Variance} = \frac{10}{5} = 2$$

Variance of Dataset B = 2

3. Calculate the Standard Deviation

Standard deviation is the square root of the variance.

Dataset A

$$SD = \sqrt{200} \approx 14.14$$

Standard Deviation = 14.14

Dataset B

$$SD = \sqrt{2} \approx 1.41$$

Standard Deviation = 1.41

4. Which Dataset Has Greater Spread?

Dataset A has a much higher variance (**200**) and standard deviation (**14.14**) than Dataset B.

Therefore, **Dataset A has the greater spread** because its values are widely distributed around the mean.

5. Which Model Appears More Consistent?

If the datasets represent model errors, **Dataset B** is more consistent.

This is because its errors are very close to the mean, resulting in a much lower standard deviation. Smaller variation indicates more stable and reliable performance.

6. Why Is Mean Alone Not Enough?

The mean only shows the average value and does not describe how the data is distributed.

In this example, both datasets have the **same mean (30)**, but Dataset A has values spread from **10 to 50**, while Dataset B has values only from **28 to 32**.

Therefore, the mean alone cannot describe the consistency or variability of the data. Variance and standard deviation are also needed for a complete comparison.

Conclusion

Both datasets have the **same mean of 30**, but their variability is very different. **Dataset A** has a **variance of 200** and a **standard deviation of 14.14**, indicating a large spread. **Dataset B** has a **variance of 2** and a **standard deviation of 1.41**, showing that its values are closely grouped around the mean. Hence, **Dataset B is more consistent**, and comparing datasets should always include both the **mean** and measures of **dispersion** such as variance and standard deviation.